Paper Insights

The Brazilian agribusiness sector is undergoing a new expansion cycle driven by sustained growth in grain production, increased vertical integration, and the reconfiguration of the Northern Arc logistics corridor. In this context, 3tentos' strategy can be understood as a structured move to capture margins across the value chain, beginning with origination, advancing into industrial processing, and culminating in the internalization of logistics infrastructure through the TUP developed in joint venture with Caramuru. Beyond scale expansion, this strategy aims to reduce vulnerabilities related to logistical bottlenecks, third-party dependence, and margin volatility. This paper examines the economic motivations, strategic mechanisms, and associated risks underlying this new cycle of growth.

New Expansion Cycle

The Brazilian agribusiness sector is undergoing a new expansion cycle driven by sustained growth in grain production, increased vertical integration, and the reconfiguration of the Northern Arc logistics corridor. In this context, 3tentos' strategy can be understood as a structured move to capture margins across the value chain, beginning with origination, advancing into industrial processing, and culminating in the internalization of logistics infrastructure through the TUP developed in joint venture with Caramuru. Beyond scale expansion, this strategy aims to reduce vulnerabilities related to logistical bottlenecks, third-party dependence, and margin volatility. This paper examines the economic motivations, strategic mechanisms, and associated risks underlying this new cycle of growth.

Main Drivers of the Expansion

The Brazilian agribusiness sector is undergoing a new expansion cycle driven by sustained growth in grain production, increased vertical integration, and the reconfiguration of the Northern Arc logistics corridor. In this context, 3tentos' strategy can be understood as a structured move to capture margins across the value chain, beginning with origination, advancing into industrial processing, and culminating in the internalization of logistics infrastructure through the TUP developed in joint venture with Caramuru. Beyond scale expansion, this strategy aims to reduce vulnerabilities related to logistical bottlenecks, third-party dependence, and margin volatility. This paper examines the economic motivations, strategic mechanisms, and associated risks underlying this new cycle of growth.

Bottlenecks Addressed by the Expansion

The development of the TUP in Pará aims to mitigate logistical bottlenecks associated with reliance on third-party port infrastructure and capacity constraints during peak harvest periods. The growth in agricultural production has increased pressure on traditional export corridors, leading to higher costs, longer lead times, and greater uncertainty in shipment scheduling. By internalizing port infrastructure, the company reduces exposure to capacity limitations, congestion, and tariff volatility, while improving coordination between origination, processing, and export activities. In this sense, the expansion directly addresses a critical friction point in the value chain: the predictability and efficiency of large-scale commodity outflows.

Expansion Implications

The company's expansion signals a structural shift in its positioning within the agribusiness value chain, reflecting a gradual transition from a predominantly commercial model toward a more integrated structure combining origination, industrial processing, and logistics. By expanding industrial capacity and strengthening its logistics infrastructure, 3tentos is expected to internalize a larger share of the grain flows originated within its own network, reducing reliance on intermediaries while capturing greater value in the processing and commercialization of agricultural commodities. This process of vertical integration may generate operational efficiency gains through better cost dilution across logistics and industrial operations, potentially supporting the expansion of operating margins over time.

At the same time, the strategy also increases the capital intensity of the business model, as industrial and logistics activities require significantly higher levels of investment in fixed assets. As a result, the expansion is expected to lead to structurally higher capex levels and a larger operational asset base, which may pressure return on invested capital (ROIC) and free cash flow generation in the short term. Additionally, the greater exposure to industrial margins and agricultural cycles increases the sensitivity of earnings to fluctuations in commodity prices and processed volumes, making capacity utilization, logistics efficiency, and capital allocation discipline key determinants for sustaining profitability over the long term.

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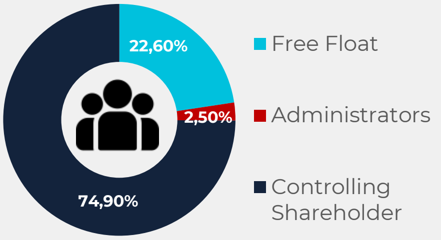
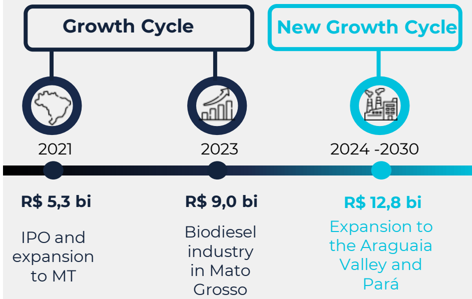
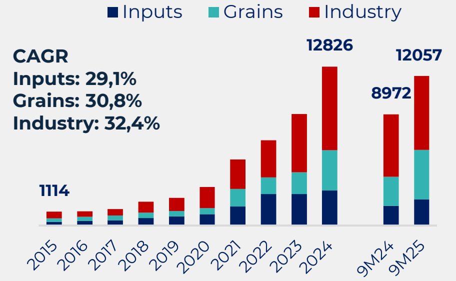
3tentos

Ticker	TTEN3
Market Cap. (R$ mn)	8,665.99
Price (R$ mn)	15.45
Price Updated	10/03/2025





Graphic 1: Revenue Segmentation(R$ Mi)



Company Operations

Origination

Origination in Brazil operates within a context of structural expansion in agricultural production. The country produced approximately 155 million tons of soybeans in the 2022/23 crop year, consolidating its position as the world's largest exporter, accounting for over 50% of global soybean trade. Total grain production (soybeans and corn) surpassed 300 million tons, with strong concentration in the Midwest region, which represents more than 45% of national output. This sustained growth increases the need for coordination across procurement, storage, and distribution, elevating the strategic role of origination companies.

However, origination in Brazil is not merely a commercial intermediation activity; it is embedded within an integrated ecosystem of inputs, credit, and producer relationships. The agricultural inputs sector is historically fragmented, capital-intensive, and highly exposed to agricultural cycles. In this environment, scale, regional capillarity, and financial discipline become key competitive advantages. 3tentos positions itself through an integrated agro-retail model, operating an expanding network of physical stores that strengthens proximity to producers, enhances recurring commercial relationships, and reduces origination risk.

The company's store expansion strategy, with ongoing openings and broader geographic coverage, deepens its presence in strategic producing regions. Each new unit not only increases input sales but also reinforces the barter model, securing future grain flows for origination. This integrated ecosystem , retail, credit, inputs, barter, and grain procurement, reduces volume volatility, improves industrial supply predictability, and builds a robust operational foundation that supports expansion into subsequent stages of the value chain, including logistics infrastructure.

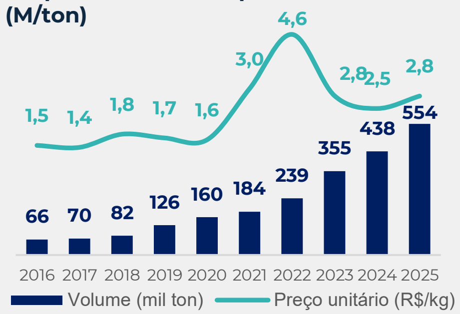
Industry

3tentos' industrial segment currently represents the company's main strategic front and the largest value-creation driver of its new expansion cycle. More than a complement to origination, industry has become the primary destination of capital allocation and the core focus of structural growth, reflecting a clear shift in the company's earnings model. In a country that produces more than 150 million tons of soybeans per year and holds over 70 million tons of installed crushing capacity, margin capture is increasingly shifting from pure grain trading toward processing. The soy complex spread (meal and oil versus beans), combined with exchange rate movements and regional logistical efficiency, underpins the structural attractiveness of crushing, particularly for vertically integrated players capable of combining proprietary origination with processing, thereby reducing basis exposure and increasing supply predictability.

At the same time, the regulatory environment reinforces the economic rationale for expansion. Brazil consumes approximately 7–8 billion liters of biodiesel per year, with blending mandates on a rising trajectory toward levels above 14–15% in the coming years. Given that roughly 70–75% of Brazilian biodiesel is produced from soybean oil, regulated domestic demand creates a structural floor for oil offtake, enhances crushing economics, and reduces exclusive dependence on export markets. This mechanism adds margin stability and greater earnings visibility to the industrial platform. In parallel, the expansion of corn ethanol strengthens the growth vector in the Center-West region. National production already exceeds 6 billion liters annually, concentrated in Mato Grosso, the state responsible for more than 30% of Brazil's corn output. The year-round operating model and the generation of co-products such as DDGS enhance asset efficiency and reinforce integration with the animal protein chain.

For 3tentos, the combination of crushing, biodiesel, and ethanol represents not merely an increase in capacity, but the consolidation of industry as the central axis of its business model, expanding margin internalization and reducing the volatility typically associated with origination. Thus, the new industrial expansion is aligned with structural drivers, global protein demand, decarbonization policies, and bioenergy growth in the Center-West, that support long-term growth and stronger operational resilience. It stands as the company's primary growth pillar, increasing value capture across the agricultural chain and consolidating 3tentos' positioning as an integrated agro-industrial platform with reduced exposure to the pure cyclicality of grain trading. Additionally, given the capital-intensive nature of crushing and biofuel plants, the industrial expansion is inherently sensitive to interest rate cycles. Higher rates increase funding costs and raise required returns; however, regulated demand in biodiesel and structural growth in corn ethanol provide greater earnings visibility, partially mitigating financial cycle volatility and reinforcing the strategic rationale for vertical integration.

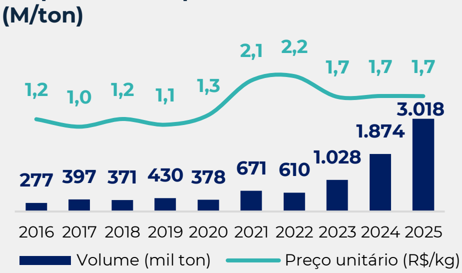
Graphic 2: Fertilizers price and volume (M/ton)



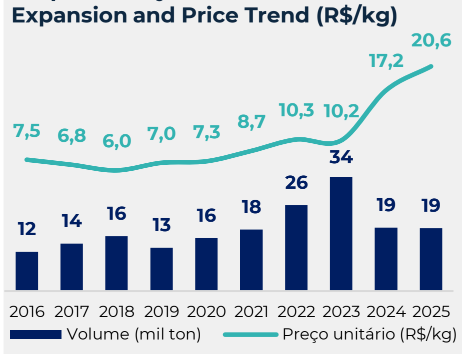
Graphic 3: Seed prices and volume (M/ton)

Map 1: Location of industrial Seed Processing Units in Pará

Graphic 4: Seed prices and volume (M/ton)



Graphic 5: Soybean Meal Volume Expansion and Price Trend (R$/kg)



Trading

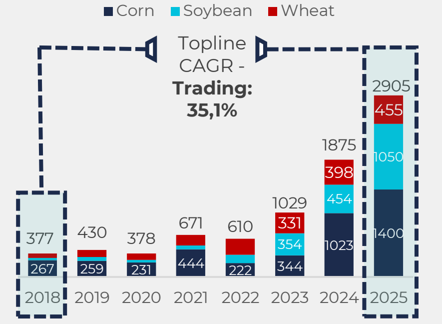
Brazil's grain trading sector has been strongly impacted by the structural growth in production, particularly in the Center-West region. Soybean output has more than doubled over the past decade, surpassing 150 million tons, while corn production has reached levels between 120 and 130 million tons. Mato Grosso, the leading producing state, accounts for more than 30% of national supply. This expansion has outpaced logistical development, increasing pressure on freight, storage, and port capacity. Historically, in record harvest years, higher shipment volumes drive up inland freight costs and increase basis volatility, compressing margins for operators reliant on third-party infrastructure.

Exchange rates represent another central driver. The strong volatility of the Brazilian real in recent years has materially affected the competitiveness of Brazilian commodities. Since soybeans and corn are priced in U.S. dollars, currency depreciation raises domestic prices and stimulates exports, while currency appreciation compresses margins and reduces export premiums. In this environment, active hedging strategies and precise commercial timing are essential to preserve trading profitability.

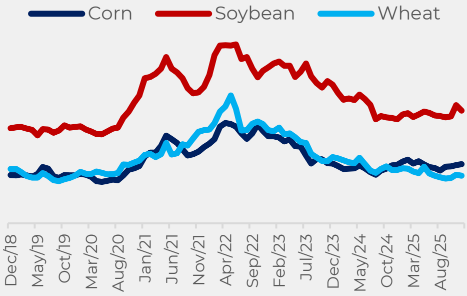
In addition, the concentration of Brazilian exports in a limited number of destinations, particularly in soybeans, increases the sensitivity of FOB premiums to shifts in demand. Fluctuations in global stock levels and purchasing pace directly affect port premiums, impacting inland netback prices. As a result, logistical efficiency becomes a critical variable: transportation costs to port can represent a significant share of product value, especially in Center-West states.

For 3tentos, these factors make logistical integration a strategic priority. Continuous production growth, combined with a historical pattern of port bottlenecks and basis volatility, reinforces the importance of controlling the export flow. The construction of a proprietary port reduces exposure to queues, demurrage, and indirect costs, improves shipment predictability, and enhances the ability to capture export premiums. In a business characterized by thin unit margins, logistical efficiency and vertical integration shift from being marginal advantages to structural determinants of competitiveness and return.

Graphic 6: 3Tentos Grain Trading Volume Expansion ('000 tons)



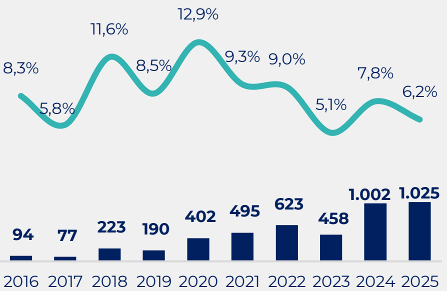
Graphic 7: Trading Grain Price Evolution: Corn, Soybean, and Wheat



Core Operational Risks

The 2025 results highlight that 3tentos has significant exposure to two central structural risks: crop failure and logistical inefficiencies. Although the year was marked by strong revenue growth (R$16.4 billion, +28.1%) and a substantial increase in grain and meal volumes traded (+43.3%), this performance itself underscores the company's dependence on volume.

Graphic 8: 3Tentos Adjusted EBITDA with hedge and Margin Evolution (R$ millions)



Crop Failure

Crop failure primarily represents a scale risk. The strong performance of the Grains segment (+60.7% annual revenue growth) was driven by higher commercialization volumes, particularly corn, supported by a record harvest in Mato Grosso. This demonstrates that a significant portion of earnings is directly anchored to physical product availability. In an adverse scenario, such as severe drought in the South or irregular rainfall in the Center-West, reduced supply would simultaneously affect origination, trading, and industrial processing. With adjusted EBITDA (including hedge effects) of approximately R$1.02 billion in 2025 and a net margin of 4.9%, the structure shows meaningful sensitivity to throughput reductions. Lower volumes would weaken fixed-cost dilution in the industrial segment, pressure origination spreads, and potentially generate underutilization of processing plants, increasing EBITDA volatility.

Table 1 – Logistics Indicators

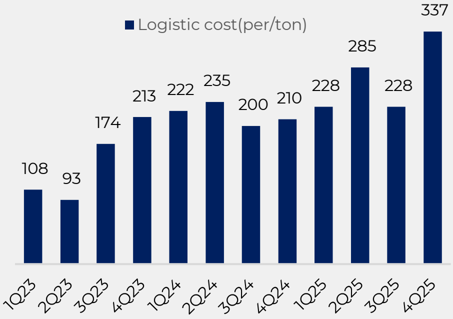
Logistics KPI	Value
Logistics expenses / Revenue	9.8%
Grain & meal volume growth	+46.9% YoY
MT share of shipments	49%

Logistical Risk

Logistical risk, in turn, manifests through cost structure and working capital dynamics. In 2025, selling, general and administrative expenses represented 13.9% of revenue in 4Q25, reflecting higher logistics costs and a greater share of volumes shipped from Mato Grosso (49% in 2025 versus 37% in 2024). The company's expansion in the Center-West increases exposure to longer and more road-dependent logistics corridors, historically characterized by freight volatility. In record harvest years, rising freight costs and potential port congestion directly impact trading netbacks and industrial margins.

Logistics costs increased sharply during the quarter, with storage and shipment expenses per ton rising about 60% year-over-year and reaching 21% of revenues from grains and soybean meal, well above the 13.6% average observed in the first nine months of the year. Management did not provide a detailed explanation, but pointed to a higher share of revenues from soybean meal and grain trading and stronger operations in Mato Grosso, which require longer transport distances. Part of the increase may also reflect structural logistics pressures, suggesting transportation costs could remain a relevant risk to margins going forward.

Graphic 9: Logistics Cost per Ton: Freight, Storage and Handling (R$/ton)



Socioeconomic context

The 3tentos–Caramuru terminal project in Miritituba (PA) represents a strategic expansion within Brazil's Arco Norte logistics corridor, improving grain export efficiency and reducing transportation costs for soy and corn flows from Mato Grosso. The development of port infrastructure and storage capacity is expected to stimulate regional economic activity through job creation, logistics services, and increased trade volumes. For both companies, the terminal strengthens vertical integration by combining 3tentos' origination scale with Caramuru's export expertise, improving supply chain control and long-term export competitiveness.

However, projects in the Pará region also carry ESG and social-impact risks, particularly given precedents such as the environmental incidents in Barcarena that heightened regulatory scrutiny and community sensitivity toward industrial projects. Large port developments in the Amazon can generate concerns related to environmental management, pressure on local infrastructure, and impacts on river-dependent communities. For investors, the key consideration is therefore the balance between the strong logistics and export upside of the project and the need for robust environmental governance and stakeholder engagement to mitigate potential regulatory or reputational risks.

Additionally, the company ended the period with net debt of approximately R$1.6 billion and a net debt/EBITDA ratio of 1.87x. While leverage remains manageable, it reinforces the compounding risk in adverse scenarios: a combination of crop failure and higher logistics costs would reduce EBITDA, mechanically increasing leverage and pressuring cash generation. In summary, recent figures demonstrate that 3tentos' business model is highly leveraged to agricultural volume and logistical efficiency. Crop failure represents a direct risk to operational scale and margin compression, while deficient logistics constitute a structural cost and execution risk. Both have the potential to materially affect cash flow generation and return on invested capital, particularly in a context of ongoing industrial expansion and increasing operational complexity.

Table 1 – Socioeconomic Indicators of Itaituba (PA)

Indicator	Value
Territorial Area	62,042.472 km²
Population (2022)	123,314 people
Expected Population (2025)	135,369 people
Schooling Rate (6–14 years)	97.4%
Gross Revenue	BRL 670.63 MN

Strategic Joint Venture Structure

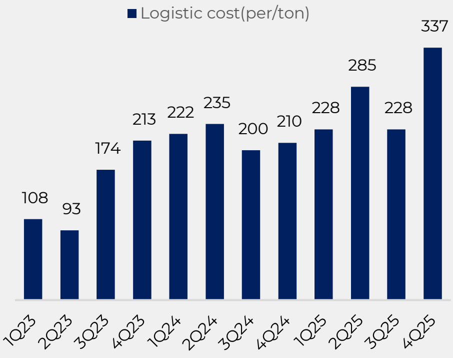
Rationale for the JV Model

Given the complexity and capital intensity associated with greenfield industrial projects in the soybean processing sector, particularly in logistics corridors still under consolidation such as the Northern Arc, a **joint venture structure** emerges as an efficient mechanism to mitigate execution risks and accelerate operational development. By sharing investments and capabilities between complementary partners, this model allows for risk dilution, greater efficiency in project implementation, and the integration of different competencies across the agribusiness value chain, from grain origination to industrial processing and the commercialization of derivatives.

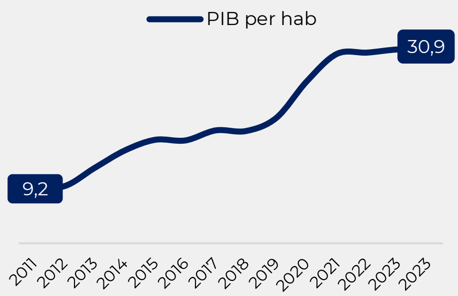
The Partnership with Caramuru

In this context, the partnership with Caramuru Alimentos proves particularly strategic. The company has extensive experience in soybean processing in Brazil, with strong industrial know-how in the production of soybean meal, oil, and biodiesel, as well as established familiarity with the logistics dynamics of the Northern Arc. Caramuru already operates a Private Use Terminal (TUP) in Miritituba, one of the main transshipment hubs within the region's export corridor, providing it with practical experience in managing grain flows and export logistics through northern ports. This combination of industrial and logistical expertise complements 3tentos' strengths in grain origination and its close relationships with producers, creating a more integrated operational structure while reducing the risks typically associated with the execution of new industrial projects.

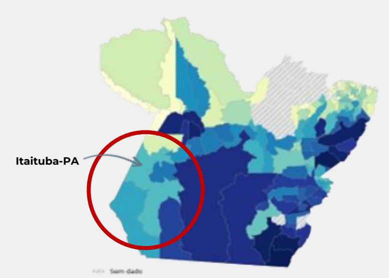
Graphic 10: Logistics Cost per Ton: Freight, Storage and Handling (R$/ton)



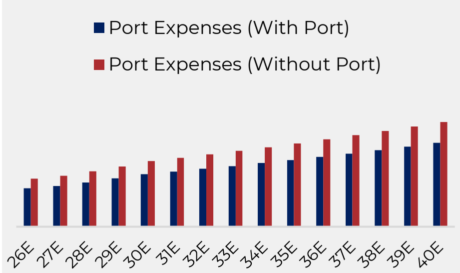
Graphic 11: Itaituba GDP per Capita Growth Driven by Northern Arc Port Expansion (R$ mil)



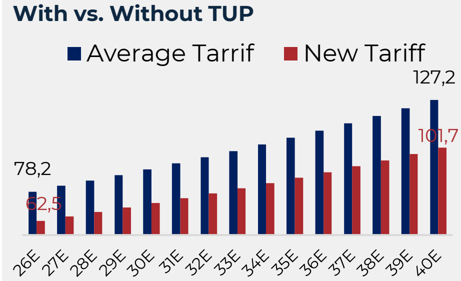
Map 2: Itaituba strategic location in between the frontiers of Pará and MT



Graphic 12: Port Expense Comparison: With vs. Without Port Access



Graphic 13: Port Tariff Comparison: With vs. Without TUP



The Northern Arc

The Northern Arc emerged as a response to the growing mismatch between the expansion of Brazil's agricultural production, particularly in the Midwest, and the historical concentration of export infrastructure in southern and southeastern ports. By bringing producing regions closer to ports located in the northern part of the country, this logistics route reduces transportation distances and mitigates reliance on traditional export corridors, contributing to greater export capacity and gradually rebalancing the relationship between the geography of production and Brazil's export infrastructure.

Agricultural Frontier Expansion

The expansion of Brazil's agricultural frontier, particularly in the Midwest and adjacent northern regions, has been reshaping the geography of grain production in the country while creating new logistics and industrial hubs along the Northern Arc corridor. States such as Mato Grosso account for a significant share of the growth in national agricultural output, driven by the availability of cultivable land, productivity gains, and the gradual expansion of transportation infrastructure toward northern export ports. In this context, establishing operations in these regions becomes strategic for companies involved in grain origination, processing, and commercialization, as proximity to producing areas reduces logistics costs, facilitates access to growing grain volumes, and improves the efficiency of export flows. For 3tentos, the decision to expand its operations toward Pará, including the construction of a new industrial facility in the region, reflects this geographical reconfiguration of Brazil's agricultural production and logistics system. By positioning industrial capacity closer to the Northern Arc export corridors, the company can shorten the distance between origination, processing, and export logistics, potentially capturing operational and logistical efficiencies across the value chain. This move also aligns with a broader industry trend, in which several players are seeking to establish a presence closer to emerging producing regions and new export routes, consolidating northern Brazil as an increasingly important vector of agribusiness expansion.

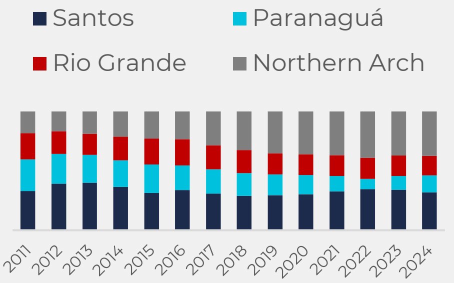
Logistics Distance and Transportation Economics

Logistics distance is one of the main determinants of the competitiveness of Brazilian agricultural exports, as transportation costs represent a significant share of the total cost of moving commodities produced inland to international markets. Historically, Brazil's logistics model has been characterized by a strong dependence on long road transportation routes between producing regions in the Midwest and export ports in the South and Southeast, such as Santos and Paranaguá. This configuration created a structural mismatch between the geography of production and the country's export infrastructure, increasing the marginal cost of transportation and exposing the agricultural supply chain to recurring logistics bottlenecks, particularly during peak harvest periods. The consolidation of export corridors along the Northern Arc partially alters this dynamic by bringing producing regions closer to export terminals, reducing overland transportation distances and enabling greater integration between logistics modes, particularly road and river transport. More than simply reducing travel distance, this reconfiguration directly affects the cost structure of the export chain, as shorter distances and improved logistics efficiency lower the marginal cost of transporting each exported ton. As a result, the reduction in logistics distance associated with the Northern Arc helps mitigate longstanding inefficiencies in Brazil's infrastructure while reinforcing the growing importance of this corridor as a strategic route for the country's agricultural exports.

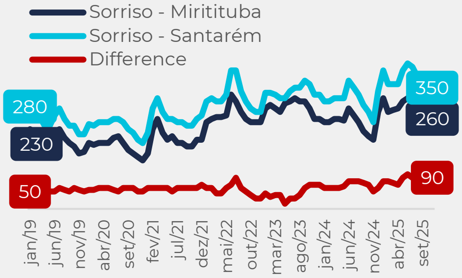
Export Routing and Strategic Market Access

Beyond logistics cost considerations, the growing relevance of the Northern Arc is also linked to the structural limitations of Brazil's traditional export corridors in the South and Southeast. Ports such as Santos and Paranaguá have historically concentrated a large share of the country's grain shipments, often operating near capacity during peak harvest periods and facing recurring congestion, logistics queues, and greater volatility in freight costs. This high level of logistics concentration creates operational risks for the export chain, as bottlenecks in these ports can disrupt the pace of grain shipments and increase uncertainty in commercial flows. In this context, the Northern Arc emerges as a strategic alternative that helps diversify export routes and reduce structural dependence on traditional corridors. By expanding the number of exit points to international markets, this logistics corridor increases the resilience of Brazil's export system while alleviating pressure on port infrastructure in the South and Southeast. Additionally, its geographic positioning allows closer access to maritime routes across the North Atlantic, facilitating trade flows toward markets such as Europe and potentially reducing sailing times compared to shipments departing from southern ports. As a result, the strengthening of the Northern Arc not only helps mitigate existing logistics bottlenecks but also enhances the competitiveness of Brazil's export chain by increasing operational flexibility, diversifying commercial routes, and reducing exposure to capacity constraints concentrated in a limited number of ports.

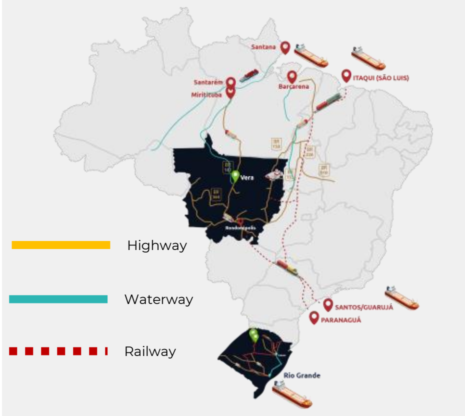
Graphic 14: Brazil Soybean Exports: Market Share By Port (BRL/ton)



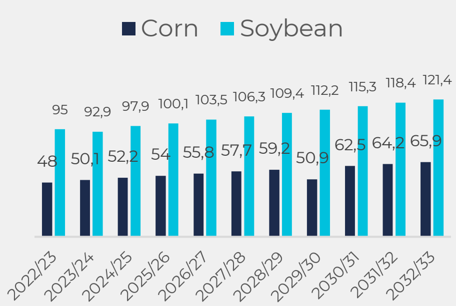
Graphic 15: Freight Comparison Departing From Sorriso (BRL/ton)



Map 3: 3tentos Logistic Ecosystem



Graphic 16: Projected Grain Export Expectations expansion (Mn/ton)



Graphic 17: Expected FCFF Terminal ramp-up drives strong long-term cash flow (Mn/ton)